1. Role of Non-Governmental Organizations (NGOs) in Disaster Management

NGOs are non-profit organisations or associations of private citizens that assist disaster-affected people. Their roles span the entire disaster cycle:

- **Emergency relief and aid distribution**: NGOs serve as providers and distributors of humanitarian aid, often reaching communities that government machinery cannot easily access.
- **Rescue and technical support**: NGOs bring technical skills important for rescue, coordination, and relief activities.
- **Rehabilitation and reconstruction**: They play a major role in long-term recovery, including construction of transit shelters and trauma counselling.
- **Preparedness and training**: NGOs assist local communities in disaster preparedness planning through training and awareness programmes.
- **Advocacy and policy engagement**: They organise platforms for advocacy, linking local communities with provincial and national governance structures.

---

2. How Early Warning Systems Function in Disaster Management

A complete early warning system is **far more than a prediction**—it is an integrated system comprising four interrelated components:

Component	Function	Key Questions
Risk Knowledge	Systematically collect data and undertake risk assessments	Are hazards and vulnerabilities known? Are risk maps available?
Monitoring & Warning Service	Develop hazard monitoring and forecasting	Are right parameters monitored? Can accurate warnings be generated?
Dissemination & Communication	Communicate risk information to those at risk	Do warnings reach all? Are they understood?
Response Capability	Build national and community response capacities	Are plans updated and tested? Are people prepared to act?

---

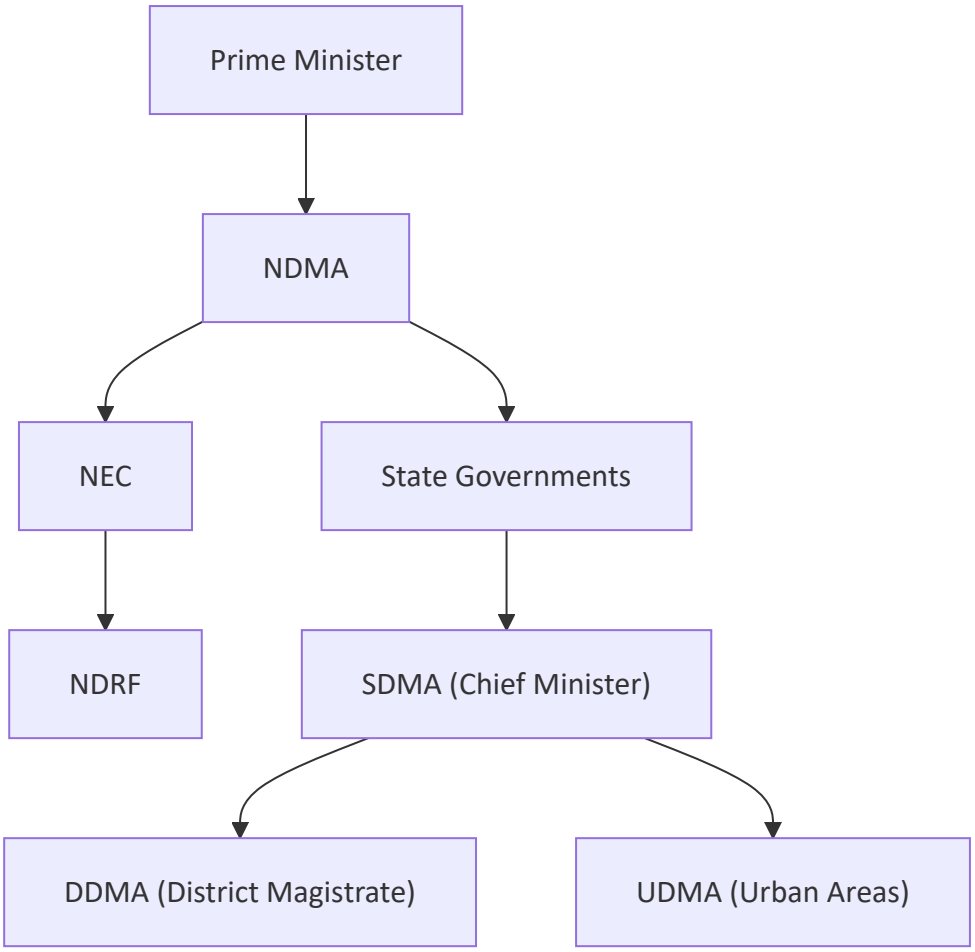
3. Government of India's Institutional Framework for Disaster Management

India's disaster management framework is established through the **Disaster Management Act, 2005** (amended in 2025). The primary responsibility for disaster management rests with **State Governments**, with the Central Government providing supplementary logistical and financial support for severe disasters.

Level	Authority	Composition
National	**NDMA** (National Disaster Management Authority)	Chaired by **Prime Minister**; lays down policies, plans, and guidelines
National	**NEC** (National Executive Committee)	Assists NDMA; coordinates and monitors implementation
State	**SDMA** (State Disaster Management Authority)	Chaired by **Chief Minister**; implements national policy at state level
District	**DDMA** (District Disaster Management Authority)	Chaired by District Magistrate/Collector; district-level planning and coordination
Response	**NDRF** (National Disaster Response Force)	16 battalions for specialised search, rescue, and response

Level	Authority	Composition
Urban	**UDMA** (Urban Disaster Management Authority)	New provision for state capitals and cities with municipal corporations

Key initiatives include the **National Disaster Management Plan** (first issued 2016, revised 2019), the **Common Alerting Protocol-based Integrated Alert System** covering all 36 States/UTs, and increased financial allocations through **SDRF, NDRF, SDMF, and NDMF**.



4. Nodal Officers: Roles and Responsibilities in Disaster Management

A **Nodal Officer** is a designated official responsible for coordinating disaster management activities within a specific jurisdiction or department. At the district level, the **Additional District Magistrate** typically serves as the Nodal Officer for Disaster Management and is in charge of the **District Emergency Operations Centre (DEOC)**.

Key responsibilities include:

- **Site inspection and immediate response**: Reach the disaster site immediately, inspect affected structures, and initiate rescue operations.
- **Coordination and reporting**: Inform higher authorities (e.g., Zonal Controlling Officer) about the magnitude of the disaster and coordinate rescue operations.
- **Resource mobilisation**: Arrange alternative accommodations and ensure readiness of response teams.
- **DEOC management**: Monitor, coordinate, and implement disaster management actions; activate Emergency Support Functions as needed.
- **Information management**: Maintain databases of disasters, losses, trained personnel, and resources; report to the Relief Commissioner's office.
- **Media coordination**: Brief media during disasters and prepare day-to-day situation reports.

5. Difference Between NIDM and NDMA

Aspect	NDMA (National Disaster Management Authority)	NIDM (National Institute of Disaster Management)
Nature	Apex **policy-making and coordinating** body	Premier **training and research** institute
Chairperson	**Prime Minister** of India	Headed by a Director
Primary Mandate	Lay down policies, plans, and guidelines for disaster management	Human resource development, training, research, and documentation in disaster management
Key Functions	Approve National Disaster Management Plan; coordinate enforcement of policies; provide funds for mitigation	Conduct training programmes; develop educational materials; undertake research; build capacity of stakeholders
Legal Basis	Established under Section 3 of DM Act 2005	Constituted under DM Act 2005; functions under Ministry of Home Affairs
Output	National policies, plans, and guidelines	Trained professionals, research reports, curricula

Detailed Functions:

NDMA Functions:

- Lay down policies on disaster management
- Approve the National Plan
- Lay down guidelines for ministries and State Authorities
- Coordinate enforcement and implementation of policy
- Arrange funds for mitigation, preparedness, and response
- Provide support to other countries affected by major disasters

NIDM Functions:

- Impart training to government officials, NGOs, and community representatives
- Conduct research on disaster risk reduction and management
- Develop educational and training materials
- Serve as a resource centre for disaster management knowledge
- Collaborate with national and international institutions for capacity building

6. Structural Mapping in Disaster Management

Structural mapping in disaster management refers to the process of identifying, documenting, and spatially representing the physical features and infrastructure of a community or region to understand vulnerabilities, capacities, and risk exposure.

How it is carried out:

1. **Participatory mapping**: Community members draw maps using local knowledge, identifying features such as houses, roads, schools, health facilities, and water sources.
2. **Feature identification**: For each mapped feature, information is recorded about **location**, **ownership**, and whether it is a **vulnerability** or a **strength**.
3. **Risk assessment**: Features are assessed against specific hazards (e.g., coastal flooding, inland flooding, wind) to determine risk levels.
4. **Transect walks**: Systematic walks through the community with key informants to observe and document spatial differences, land use

zones, and danger areas.

5. **Integration with GIS**: Modern structural mapping uses Geographic Information Systems to layer hazard data, infrastructure data, and demographic information for spatial analysis.

---

7. Mitigation Dos and Don'ts for Cyclones, Hurricanes, and Tornadoes

Before (Watch Area):

- **DO** frequently listen to radio, TV, and NOAA Weather Radio for official bulletins
- **DO** fuel and service family vehicles; have extra cash on hand
- **DO** stock up on canned food, first aid supplies, drinking water, and medications
- **DO** prepare to cover windows and doors with shutters or shielding materials
- **DO** bring inside lawn furniture and loose lightweight objects

Before (Warning Area):

- **DO** complete preparation activities (storm shutters, store loose objects)
- **DO** follow instructions issued by local officials; leave immediately if told to do so
- **DO** leave mobile homes in any case
- **DO** notify neighbors and a family member outside the warned area of evacuation plans

During (If Staying Home):

- **DO** stay inside a well-constructed building
- **DO** turn refrigerator to maximum cold and open only when necessary
- **DO** fill bathtub and large containers with water for sanitary purposes
- **DO** take refuge in a small interior room, closet, or hallway away from windows
- **DO NOT** use elevators
- **DO NOT** go outside during the storm

During (Tornado Specifically):

- **DO** go to basement or small, interior, windowless room on lowest level
- **DO** protect head and neck with helmets, pillows, or blankets
- **DO NOT** stay in a mobile home
- **DO NOT** try to outrun a tornado in a vehicle
- **DO NOT** shelter under an overpass

After the Storm:

- **DO** wait until area is declared safe before entering
- **DO** avoid weakened bridges and washed-out roads
- **DO NOT** drive into flooded areas
- **DO NOT** drink or prepare food with tap water until certain it is not contaminated
- **DO NOT** use candles or open flames indoors; use a flashlight

---

8. Role of the National Disaster Response Force (NDRF)

The **NDRF** was raised on **19 January 2006** under the Disaster Management Act, 2005, and has grown from 8 to **16 battalions**. It is a

specialised force mandated to respond to **all natural and man-made disasters** except fires.

Key Roles and Capabilities:

- **Collapsed Structure Search & Rescue (CSSR)**: Earthquakes, building collapses, landslides
- **Flood Water Rescue (FWR)**: Drowning incidents, boat capsizes
- **Chemical, Biological, Radiological & Nuclear (CBRN)**: Specialised response to hazardous material incidents
- **Mountain & Rope Rescue**: Operations in difficult terrain
- **Medical First Response (MFR)** and **Animal Rescue**
- **Forest Fire Response**: New mandate added in 2022

Operational Achievements:

- Conducted more than **12,000 operations**, rescued over **1,58,000 persons**, and evacuated more than **8.5 lakh persons**
- International response: Japan Triple Disaster (2011), Nepal Earthquake (2015), Turkey Earthquake (2023)
- Pre-positioning during monsoon: 107 teams earmarked for pre-positioning in 2024

---

9. Difference Between Structural and Non-Structural Mitigation Measures

Aspect	Structural Mitigation	Non-Structural Mitigation
Definition	Physical construction or engineering techniques to reduce hazard impacts	Measures not involving physical construction; use knowledge, practice, or agreement
Focus	Changes to the **built environment**	Changes to **human behaviour** and **policies**
Examples	Dams, levees, flood walls, earthquake-resistant buildings, evacuation shelters, ocean wave barriers	Building codes, land-use planning laws, public awareness programmes, training, insurance, research and assessment
Approach	Engineering and technology-based	Policy, education, and behaviour-based
Timeframe	Often long-term, capital-intensive	Can be implemented more quickly and at lower cost

---

10. Role of the Internet of Things (IoT) in Disaster Management

The **Internet of Things (IoT)** refers to networks of interconnected sensors and devices that collect and transmit real-time data. In disaster management, IoT enables:

- **Real-time monitoring**: Sensors and Wireless Sensor Networks (WSN) provide initial detection of disasters and continuous real-time data on flood levels, seismic activity, fire, and weather conditions.
- **Early warning systems**: IoT applications provide early warning in urban spaces during floods, earthquakes, and fires, allowing search and rescue teams to respond within the **golden response time**.
- **Communication in damaged areas**: Technologies like **LoRaWAN** (Long Range Wide Area Network) and cloud/fog computing maintain connectivity in remote or damaged areas.
- **Data-driven decision-making**: IoT data enables real-time monitoring and decision execution for more effective response.

---

11. Role of Geographic Information Systems (GIS) in Disaster Management

GIS is a "location science" that allows people to **visualise what is happening and where**, placing information in a spatial framework on a map. Its roles include:

- **Hazard and risk mapping**: Identifying areas prone to flooding, earthquakes, landslides, and other hazards.
- **Vulnerability analysis**: Integrating demographic and socio-economic data to identify at-risk populations and communities.
- **Evacuation planning**: Determining evacuation routes, safe areas, and shelter locations.
- **Resource allocation**: Optimising the placement of emergency resources and personnel based on spatial risk assessment.
- **Real-time situational awareness**: During disasters, GIS provides updated maps of flood impact zones, critical facilities, and affected communities.
- **Post-disaster recovery**: Supporting damage assessment and reconstruction planning.

---

12. Key Provisions of the Disaster Management Act, 2005 (as amended 2025)

The **Disaster Management Act, 2005** provides the legal and institutional framework for disaster management in India.

Key Provisions:

- **Establishment of Authorities**:
  - **NDMA**: Chaired by Prime Minister; lays down policies and plans
  - **SDMA**: Chaired by Chief Minister in each state
  - **DDMA**: Chaired by District Magistrate at district level
- **National Executive Committee (NEC)**: Assists NDMA; prepares National Plan; coordinates and monitors implementation
- **National Disaster Response Force (NDRF)**: Specialised response force under DM Act
- **National Disaster Management Plan**: First issued 2016, revised 2019; defines roles of all sectors, ministries, and departments

2025 Amendment Key Changes:

- Strengthened roles of NDMA, SDMAs, and DDMAs in risk assessment, technical assistance, and multi-hazard planning
- Statutory status to **High-Level Committee (HLC)** and **National Crisis Management Committee (NCMC)**
- Provision for **Urban Disaster Management Authority (UDMA)** in state capitals and cities with municipal corporations
- Recognition of risks from **extreme climate events**
- Improved functioning of NEC, SEC, and DDMAs as planning and execution arms
- Enhanced financial mechanisms and accountability structures

---

13. Dos and Don'ts for Major Disaster Types

Earthquake:

- **DO** DROP to the ground, take COVER under sturdy furniture, and HOLD ON until shaking stops
- **DO** stay away from windows, heavy furniture, and hanging objects
- **DO** use a whistle or tap on a pipe if trapped
- **DO NOT** use elevators
- **DO NOT** move from one place to another during severe shaking
- **DO NOT** light matches or kick up dust

Flood:

- **DO** move to higher ground immediately
- **DO** use a pole or stick to test ground while walking through flooded areas
- **DO NOT** walk through flowing water—six inches can knock you off your feet

- **DO NOT** drive through flooded areas—more people drown in cars than anywhere else
- **DO NOT** drive around road barriers

Fire:

- **DO** remain calm, unplug electrical appliances, and keep buckets of water and blankets ready
- **DO** learn at least two escape routes and keep them free from obstacles
- **DO** Stop, Drop, and Roll if clothes catch fire
- **DO NOT** burn crackers in crowded or congested places
- **DO NOT** apply adhesive dressing on burnt areas
- **DO NOT** throw lighted cigarette butts

Landslide:

- **DO** stay alert and awake—many fatalities occur during sleep
- **DO** listen for unusual sounds (trees cracking, boulders knocking)
- **DO** move away from the landslide path as quickly as possible
- **DO** avoid river valleys and low-lying areas
- **DO NOT** build houses near steep slopes, mountain edges, or drainage ways

Tsunami:

- **DO** treat earthquake or strong ground rumbling as a warning signal
- **DO** climb to a raised platform or highest floor of a building
- **DO** evacuate immediately if you hear an official warning
- **DO NOT** go down near the beach to watch the tsunami
- **DO NOT** return home until authorities advise it is safe

Cyclone/Hurricane:

- **DO** follow evacuation orders immediately
- **DO** shelter in an interior room away from windows if staying
- **DO** leave mobile homes in any case
- **DO NOT** venture out during the storm
- **DO NOT** take shelter in small sheds or under isolated trees